Momentum in two dimensions

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Conserve momentum separately along perpendicular axes.

SUCCESS CRITERIA - I can...

1. Choose and label two axes.
2. Add momentum components along matching directions.
3. Connect component sums to the final motion.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Two carts approach: one has momentum 5 kg·m/s east and the other 3 kg·m/s west. Explain why adding their magnitudes gives the wrong total; find the signed total.

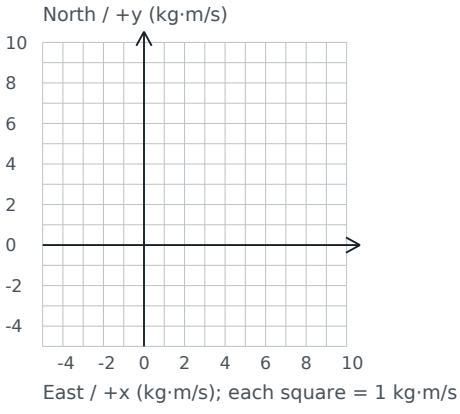
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $p_{x,i} = p_{x,f}$; $p_{y,i} = p_{y,f}$; $p_x = mv_x$; $p_y = mv_y$. Use total system momentum in each direction; negligible external impulse. CPA components are supplied.

Worked reasoning: Two objects have momenta (3, 0) and (0, 4) kg·m/s. Their total is (3, 4), magnitude 5 kg·m/s. Add components before finding the magnitude.

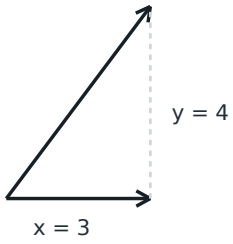
Your turn: Draw a tip-to -tail momentum diagram and label each axis with units.



Marzano check - circle: ① ② ③

03 Read and interpret the model

Two incoming arrows point east and north. Sketch the direction of their vector sum. Can their ordinary lengths be added for the diagonal magnitude?

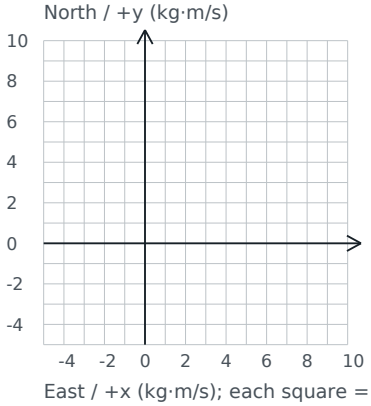


The arrows represent horizontal and vertical components; use their directions to describe the sum.

Marzano check - circle: ① ② ③

04 Apply the physics

Incoming total momentum is 6 kg·m/s east and 8 kg·m/s north. Two objects stick with total mass 2 kg. Divide each component by 2 to find velocity components.



Marzano check - circle: ① ② ③

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Continue down each column.

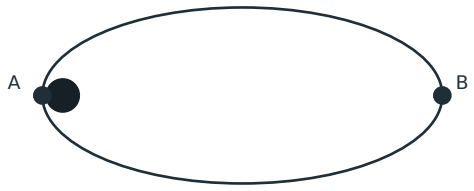
05 Explain your reasoning

Why must both momentum component sums use the same system and same time interval?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

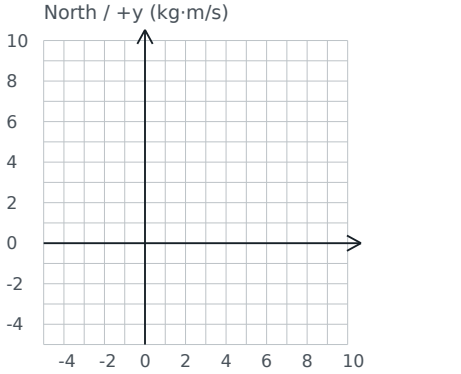
An ideal asteroid orbit has nearest distance 4 AU and farthest distance 8 AU. Where is it fastest? Explain with the equal-area law.
Label the orbital features used in your explanation.



Marzano check - circle: 1 2 3

07 Exit check

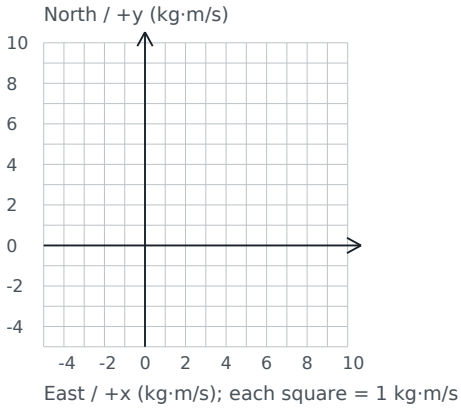
Total momentum is 4 kg·m/s west and 2 kg·m/s south. Combined mass 2 kg. Give final velocity components.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

A joined pair of total mass 2 kg has total momentum 6 kg·m/s east and 8 kg·m/s north. Find its velocity components using each momentum component divided by 2.



Marzano check - circle: 1 2 3